Finite structure in a min–max quadratic sign problem

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Abstract

For a symmetric zero-diagonal sign matrix, let $M(A)$ be the largest absolute value of its quadratic form on the Boolean cube, and put $F(n) = \min_A M(A)$. We give an auditable account of the exact values through order fourteen, including explicit computational trust boundaries. We also prove an exact energy-weighted covering formula for one-vertex extension, record finite non-heredity of optimizers, and give a sharp completion theorem for one fixed oriented-Hadamard frame which implies $F(16) \leq 30$. The asymptotic question whether $F(n)/n^{3/2}$ converges remains open.

Status. This is a focused manuscript draft. Every unconditional analytic statement is proved below. Claims labelled computer-assisted depend on the stated enumeration boundary. No result here proves convergence or nonconvergence.

1 Problem and elementary structure

Let $A = (a_{ij})$ range over symmetric zero-diagonal matrices with $a_{ij} \in \{-1, 1\}$. Define

$$Q_A(x) = \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j, \qquad M(A) = \max_{x \in \{-1,1\}^n} |Q_A(x)|, \qquad F(n) = \min_A M(A).$$

Switching by $z \in \{-1, 1\}^n$ sends A to $D_z A D_z$ and leaves $M(A)$ unchanged. Vertex permutation and global negation also preserve $M(A)$.

Lemma 1.1 (Puncturing and parity)**.** *For every $n \geq 2$,*

$$F(n) \leq F(n+1) \leq F(n) + n, \qquad F(n) \equiv \binom{n}{2} \pmod 2,$$

and consequently

$$F(n) \geq F(n-1) + ((n-1) \bmod 2).$$

Proof. Restrict an order-$(n+1)$ signing to its first n vertices. For every old spin x, its restricted energy is the average of the two full energies obtained by setting the new spin to 1 and -1. This gives $F(n) \leq F(n+1)$. Extending an optimal order-n signing by arbitrary incident signs changes every energy by at most n, proving the upper bound. Every energy is a sum of $\binom{n}{2}$ signs and therefore has that parity; the maximum and its minimum over signings have the same parity. When $n-1$ is odd, the parities at consecutive orders differ, so monotonicity improves by one. $\square$

There is also a useful exact coding formulation. Put $E_n = \binom{n}{2}$ and

$$D_n = \{(x_i x_j)_{i<j} : x \in \{-1,1\}^n\} \cup \{-(x_i x_j)_{i<j} : x \in \{-1,1\}^n\}.$$

If $\rho(D_n)$ denotes its covering radius in the Hamming cube, then

$$F(n) = E_n - 2\rho(D_n). \tag{1}$$

Indeed, correlation with a sign word is $E_n - 2$ times Hamming distance, and adjoining the negatives converts absolute correlation to nearest-codeword distance. This identity is exact but does not by itself control the second-order deficit of size $n^{3/2}$.

2 Exact values through order fourteen

Theorem 2.1 (Exact finite sequence)**.** *The exact values from order two through order fourteen are*

n	2	3	4	5	6	7	8	9	10	11	12	13	14
$F(n)$	1	3	4	4	5	9	10	12	13	17	18	20	21.

The lower bounds at orders eleven and thirteen are computer-assisted with the nauty completeness boundary described below.

Certificate architecture. For orders at most ten, switching makes every edge incident with a chosen root positive. The remaining signing is an ordinary graph on $n-1$ vertices. Isomorph-free generation, exact Gray-code energy evaluation, and independent decoding reproduce

$$1, 3, 4, 4, 5, 9, 10, 12, 13.$$

All energy and threshold decisions use integers.

For order eleven, the certificate scans all 12,005,168 unlabeled graphs on ten residual vertices. Exact degree and energy filters leave 2,153,606 eligible classes; none admits maximum at most fifteen. An explicit order-eleven signing has maximum seventeen. Parity excludes sixteen, hence $F(11) = 17$. The independently checked one-vertex extension then gives an order-twelve signing with maximum eighteen. Puncturing and parity give $F(12) \geq 18$.

For order thirteen, eight fail-closed nauty shards together contain exactly 1,018,997,864 unlabeled graphs on eleven residual vertices. The certificate pins each producer's exit status, byte count, record count, and SHA-256 stream digest. Exactly two rooted records survive the order-twelve threshold; direct evaluation gives internal maximum eighteen and minimum one-vertex extension value twenty-four for both. Therefore an order-thirteen signing of maximum eighteen cannot exist. Parity excludes nineteen. Every thirteen-vertex principal submatrix of the order-fourteen Paley conference matrix has maximum twenty, so $F(13) = 20$. Monotonicity and parity give $F(14) \geq 21$, while direct evaluation of that conference matrix gives maximum twenty-one.

The exhaustive lower certificates trust nauty's completeness and canonical isomorph-free generation. They do not infer infeasibility from a solver timeout. The explicit upper witnesses and all their complete energy scans are independently reproducible without that trust boundary. □

Remark 2.2. The order-thirteen scan finds two *rooted records*, not a theorem that there are only two unrooted optimal switching classes. Likewise, the certificate proves that the order-fourteen conference matrix is optimal but does not by itself prove uniqueness.

3 The exact one-vertex state

For an order-n signing B, let $E(B)$ be the smallest maximum obtainable by adjoining one vertex with arbitrary incident sign vector. Write

$$\mathbb{P}_n = \{-1,1\}^n/\{x \sim -x\}, \qquad d_\pm([u],[v]) = \min\{d_H(u,v), d_H(u,-v)\}.$$

Put $M = M(B)$ and define

$$w_B([x]) = \frac{M - |Q_B(x)|}{2}, \qquad \rho_{\mathrm{w}}(B) = \max_{[b]\in\mathbb{P}_n} \min_{[x]\in\mathbb{P}_n} \big(d_\pm([b],[x]) + w_B([x])\big).$$

The weights are nonnegative integers because all order-n energies have the same parity.

Theorem 3.1 (Weighted Bellman identity)**.** *For every order-n signing B,*

$$E(B) = M(B) + n - 2\rho_{\mathrm{w}}(B). \tag{2}$$

Equivalently, with $\delta_{\mathrm{w}}(B) = \lfloor n/2 \rfloor - \rho_{\mathrm{w}}(B)$,

$$F(n+1) = \min_B \big(M(B) + (n \bmod 2) + 2\delta_{\mathrm{w}}(B)\big). \tag{3}$$

Only configurations satisfying $|Q_B(x)| \geq M(B) - 2\lfloor n/2 \rfloor$ can affect the inner covering minimum.

Proof. Let $b \in \{-1,1\}^n$ be the incident signs and t the new spin. Pairing the two values of t gives

$$\max_{t\in\{-1,1\}} |Q_B(x) + t\, b\cdot x| = |Q_B(x)| + |b \cdot x|.$$

Since $|b \cdot x| = n - 2d_\pm([b],[x])$, the maximum over old spins for fixed b equals

$$M + n - 2 \min_{[x]\in\mathbb{P}_n} \big(w_B([x]) + d_\pm([b],[x])\big).$$

Minimizing over $[b]$ proves (2). Minimizing over B and using $n - 2\lfloor n/2 \rfloor = n \bmod 2$ proves (3). If $w_B([x]) > \lfloor n/2 \rfloor$, that state loses to an exact maximizer at projective distance at most $\lfloor n/2 \rfloor$, proving the final claim. $\square$

The formula explains why exact optimizers need not be valid predecessors. It also shows why a scalar energy is not a closed state: the spatial placement of every energy layer in the projective cube matters.

4 Finite optimizer non-heredity

For fixed internal blocks A and B, define

$$J(A,B) = \min_{C\in\{-1,1\}^{s\times k}} M \begin{pmatrix} A & C \\ C^\mathsf{T} & B \end{pmatrix}.$$

Proposition 4.1 (A complete $2+8$ obstruction)**.** *Among optimal blocks of orders two and eight,*

$$\min_{M(A)=F(2),\, M(B)=F(8)} J(A,B) = 15 > F(10) = 13.$$

Consequently no optimal order-ten signing contains an optimal order-eight principal submatrix. The smallest internal maximum among order-eight blocks admitting a $2+8$ completion of maximum thirteen is twelve.

This is a finite exhaustive proposition. The verifier enumerates every relevant switching-permutation class, every rectangular cross block modulo the valid symmetries, and every projective full spin. It establishes an "internal sacrifice" phenomenon: minimizing the blocks separately can make their joint optimum worse.

There is a second exact failure of scalar state. Two optimal order-nine classes have the same complete projective absolute-energy histogram

$$\#\{|Q_B| = 0, 4, 8, 12\} = (60, 111, 60, 25),$$

so all their scalar partition functions agree at every temperature, but their minimum one-vertex extension values are thirteen and fifteen. Their weighted covering geometry, not their energy multiset, distinguishes them.

5 A sharp oriented-Hadamard completion

Let

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \end{pmatrix}, \quad (a_{01}, a_{02}, a_{03}, a_{12}, a_{13}, a_{23}) = (1, 1, 1, -1, 1, -1).$$

For arbitrary symmetric zero-diagonal order-four signings D_0, D_1, D_2, D_3, let $L(D_0, D_1, D_2, D_3)$ have diagonal blocks D_i, upper cross block ij equal to $a_{ij}H$, and transpose blocks below.

Theorem 5.1 (Sharp completion of the pinned frame)**.**

$$\min_{D_0, D_1, D_2, D_3} M(L(D_0, D_1, D_2, D_3)) = 30. \tag{4}$$

In particular, $F(16) \le 30$.

Proof. Put

$$u_0 = (1, 1, 1, 1), \quad u_1 = (1, 1, 1, -1), \quad u_2 = (1, 1, -1, 1), \quad u_3 = (1, 1, -1, -1).$$

Direct multiplication gives the following cross-only energies:

	(x_0, x_1, x_2, x_3)	Q_{cross}
1+	$(u_0, u_0, -u_0, u_1)$	28
1−	$(u_0, -u_3, -u_0, -u_1)$	−28
2+	$(u_3, u_3, -u_3, u_2)$	28
2−	$(u_3, -u_2, -u_3, -u_2)$	−28
3+	$(u_3, u_2, -u_3, u_2)$	28
3−	$(u_3, -u_0, -u_3, -u_2)$	−28.

Within each pair the projective spins on clouds zero, two, and three agree, so their internal contributions cancel in the comparison. If the full maximum were at most twenty-eight, the three comparisons would force

$$q_{D_1}(u_0) \le q_{D_1}(u_3) \le q_{D_1}(u_2) \le q_{D_1}(u_0).$$

All three values would be equal. But

$$q_{D_1}(u_0) - q_{D_1}(u_2) = 2(d_{13} + d_{23} + d_{34}) \ne 0,$$

because a sum of three signs cannot vanish. Order-sixteen energies are even, so the restricted minimum is at least thirty.

For attainment, list the six internal edges in order $01, 02, 03, 12, 13, 23$ and take

$$P = (1, 1, 1, -1, 1, -1), \qquad R = (-1, -1, 1, 1, 1, -1), \qquad (D_0, D_1, D_2, D_3) = (P, R, P, R).$$

An exhaustive exact scan of all 2^{15} projective spins gives maximum thirty. Independent direct and blockwise implementations reproduce the complete energy histogram. □

Both P and R have maximum four and are switching-permutation equivalent. Nevertheless their ordered response vectors to this frame are

$$(2, 0, 4, -2, 0, 2, -2, -4), \qquad (0, -2, 2, -4, -2, 0, 4, 2).$$

Using one literal common internal block in all four clouds has minimum thirty-eight. Thus neither $M(D)$ nor the unframed switching class is a closed substitution state; orientation relative to the cross frame is essential. The theorem is restricted to one frame. It neither proves $F(16) = 30$ nor supplies an iterable amplification law.

6 Reproducibility and scope

The repository contains source-built verifiers for all statements above. The principal commands are:

```
python3 verification/research_exact_small_n.py --max-n 10
python3 verification/research_order11_certify.py
python3 verification/research_order13_certify.py --full-stream --jobs 8
python3 verification/research_cross_block_composition.py
python3 verification/verify_nonlinear_bellman.py
python3 verification/verify_framed_hadamard_lift_30.py
```

The order-eleven and order-thirteen complete lower scans require nauty and are much more expensive than witness verification. Exact commands, pinned counts, expected outputs, corruption controls, and the producer trust boundary are recorded in `verification/README.md`.

These finite results identify optimizer-sensitive composition as the main structural difficulty. They do not determine an asymptotic constant and do not constitute an answer to the MathOverflow limit question. The companion manuscript `composition_framework.tex` isolates the remaining cross-order theorem.

References

[1] P. Ivanisvili, *Min-max of a quadratic form of plus-minus ones*, MathOverflow question 413935, 2022.

[2] B. D. McKay and A. Piperno, *Practical graph isomorphism, II*, Journal of Symbolic Computation 60 (2014), 94–112.